



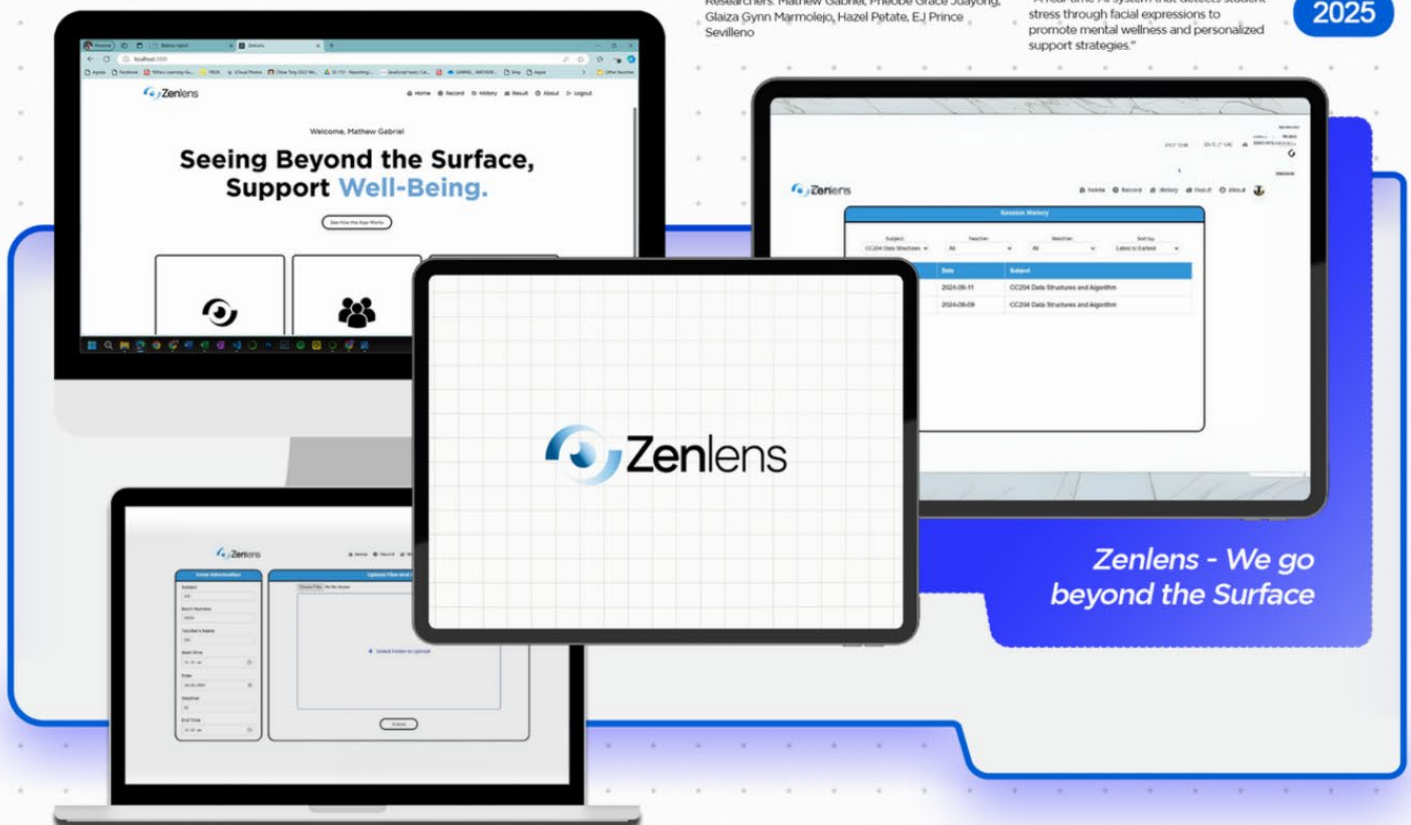
USER MANUAL

ZENLENS: IMAGE-BASED STRESS DETECTION AND TAILORED RECOMMENDATIONS FOR ENHANCED STUDENT STRESS MANAGEMENT

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"A real-time AI system that detects student stress through facial expressions to promote mental wellness and personalized support strategies."

2025



BS Computer Science Major in AI | Group 4

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ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced
Student Stress Management

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Computer Science

by

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Disclaimer

This research project and its corresponding documentation entitled “ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management” is submitted to the College of Information and Communications Technology, West Visayas State University, in partial fulfillment of the requirements for the degree, Bachelor of Science in Computer Science. It is the product of our own work, except where indicated text.

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Getting Started

Introduction

College students frequently experience academic stress due to factors such as heavy workloads, challenging coursework, and low motivation. These stressors often contribute to mild but persistent emotional strain. To aid educators in managing their students' mental wellbeing, the present study aims to develop a system that monitors students mental stress by analyzing their facial expressions using Computer Vision techniques, specifically through a Facial Expression Recognition (FER) system. The system is designed to detect and classify various facial expressions, which are then aggregated on a weekly basis to generate an emotional overview, providing insights into students' mental wellbeing over time.

This User Manual contains all essential information for the user to make full use of the Zenlens Facial Expression Recognition System. Graphics and pictures are also provided for the step-by-step instructions.

System Requirements

Windows

Minimum system requirements:

Operating System (OS):	Windows 10 64-bit: Home or Pro (build 19045 or later)
Processor:	64-bit processor with Second Level Address Translation (SLAT)
Memory:	8GB RAM

Recommended Specs:

Processor:	Intel(R) Core (TM) i5-10300H @ 2.5GHz or Higher
Memory:	8GB RAM or Higher

Installation

Before installing the system package, ensure that you have the following:

- **VSCode** or any IDE (integrated development environment) suitable or alternative.
- **Python** is installed in your computer.
- **Google Chrome** or any up-to-date web browser.

Installing Node Js

1. Download Node.js Installer

In your preferred web browser, navigate to <https://nodejs.org/en/download>. Click on the Windows Installer (.msi) button to download the latest version.

At the time of writing, version v24.1.0 was the latest version. The version used for the study was v20.7.0. The Node.js installer includes the NPM package manager (which is also crucial for later steps)



The screenshot shows the 'Download Node.js®' page. At the top, there are dropdown menus for 'v24.1.0 (Current)', 'for Windows', 'using Docker', and 'with npm'. Below these is a dark box containing Docker installation instructions in a PowerShell terminal format. At the bottom, there are two green buttons: 'Windows Installer (.msi)' and 'Standalone Binary (.zip)'. A blue arrow points to the 'Windows Installer (.msi)' button.

```
1 # Docker has specific installation instructions for each operating system.
2 # Please refer to the official documentation at https://docker.com/get-started/
3
4 # Pull the Node.js Docker image:
5 docker pull node:24-alpine
6
7 # Create a Node.js container and start a Shell session:
8 docker run -it --rm --entrypoint sh node:24-alpine
9
10 # Verify the Node.js version:
11 node -v # Should print "v24.1.0".
12
13 # Verify npm version:
14 npm -v # Should print "11.3.0".
```

PowerShell Copy to clipboard

Docker is a containerization platform. If you encounter any issues please visit [Docker's website](https://docker.com)

Or get a prebuilt Node.js® for Windows running a x64 architecture.

Windows Installer (.msi) Standalone Binary (.zip)

2. Install Node.js and NPM from Browser

- 1) Once the installer finishes downloading, launch it. Open the downloads link in your browser and click the file. Or browse to the location of the saved file and double-click to launch.
- 2) The system will ask if you want to run the software – click Run.

- 3) You will be welcomed to the Node.js Setup Wizard – click Next.
- 4) On the next screen, review the license agreement. Click Next if you agree to the terms and install the software.
- 5) The installer will prompt you for the installation location. Leave the default location, unless you have a specific need to install it somewhere else then – click Next.
- 6) The wizard will let you select components to include or remove from the installation. Again, unless you have a specific need, accept the defaults by clicking Next.
- 7) Finally, click the Install button to run the installer. When it finishes, click Finish.

3. *Verify Installation*

- 1) After installation, open command prompt (or PowerShell), and enter the following:
`node -v`
- 2) The system should display the Node.js version installed on your system. You can do the same for the NPM using the following code on the command prompt (or PowerShell):
`npm -v`
- 3) Once both versions are shown, then you will have successfully installed Node JS and NPM.

Cloning the Zenlens System from the GitHub Repository

1. Open your terminal in your VSCode (or preferred IDE) or your command prompt.
2. Enter the following command:

```
git clone https://github.com/epsevilleno/thesis-zenlens-g4cs4a.git
```

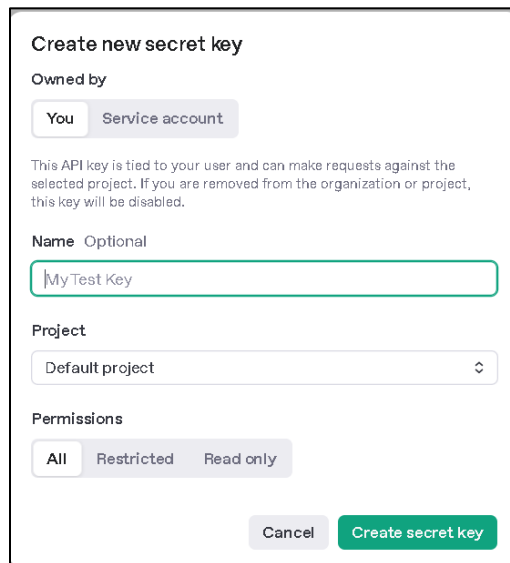
Note: Cloning a repository needs a constant internet connection. This may take several minutes to do so.

Download the ZFNet Model

1. Click [here](#) to download the ZFNet Model used in the study or go to <https://drive.google.com/file/d/1T5cWlb4hEEG46SSDZI29OuZDhycz5nkB/view>
2. You would be redirected to a Google Drive link where you can click the download button.
3. Take note of the model's file location as this is crucial for the next steps.

Get your OpenAI API Key

1. Head over to <https://platform.openai.com/api-keys> to get your API Key (Make an account if you don't have one yet).
2. Click on the "Create a new secret key" button found on the right side of the screen.
3. You will be greeted with this pop-up where you can enter a name (optional). After that click the "Create secret key" button



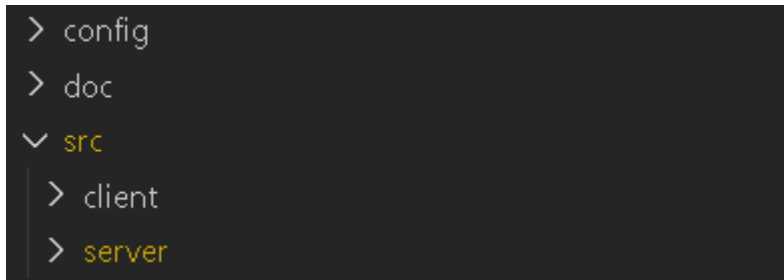
The screenshot shows a modal window titled "Create new secret key". It has two tabs: "You" (selected) and "Service account". Below the tabs is a warning: "This API key is tied to your user and can make requests against the selected project. If you are removed from the organization or project, this key will be disabled." There is a text input field for "Name" with the placeholder "Optional" and the value "My Test Key". Below that is a "Project" dropdown menu showing "Default project". At the bottom, there are "Permissions" tabs: "All" (selected), "Restricted", and "Read only". At the very bottom are two buttons: "Cancel" and "Create secret key".

4. A moment will pass and once a secret key was generated, copy it and save your key in a secure and accessible location. We will use it later.

Note: Please save your secret key in a safe place since you won't be able to view it again. Keep it secure, as anyone with your API key can make requests on your behalf. If you do lose it, you'll need to generate a new one.

Setting up the backend

1. Open the cloned repository in VSCode (or an IDE of your choice). You will see three main folders: config, doc, and src. In the src folder, you will see two folders named (1) client and (2) server. The Client folder contains the frontend of the web application, while the Server folder contains the backend side.



2. Open the server folder and put the recently downloaded model inside it.
3. After that, create a `.env` file inside the config folder. Create a variable named `OPENAI_API_KEY` and paste the secret key you created a while ago.
4. After that, open a terminal and make sure you are in the project root directory.
5. Then navigate to the server directory by running the following command:

```
cd src/server
```

6. Once you are in the server directory, create a virtual environment by running this command:

```
Python -m venv venv
```

7. This will create a virtual environment named 'venv'. Depending on your computer specifications, it will take a minute or two.

Note: If you prefer another name, you can modify the second venv in the command above.

8. Still, at the server directory. After creating a virtual environment, activate it by running the following command:

```
venv\scripts\activate
```

Note: You'll notice that once the virtual environment is activated, the name of your virtual environment appears before the working directory (often enclosed in parentheses).

9. Now, that you are inside the virtual environment, Install the necessary packages in the requirements.txt by running the command:

```
pip install -r requirements.txt or
```

```
python -m pip install -r requirements.txt
```

10. After installing, you can now run the following command (still inside the virtual environment):

```
Python app.py
```

Note: You have the option to do the step 8 after setting up the frontend

Setting up the frontend

1. Open a different terminal (If you are running the backend) with your root directory in the prompt.
2. Navigate to the client directory by running the command:

```
cd src/client
```

3. Run the following command to install the necessary dependencies:

```
npm install
```

4. To start the development server, run the following command:

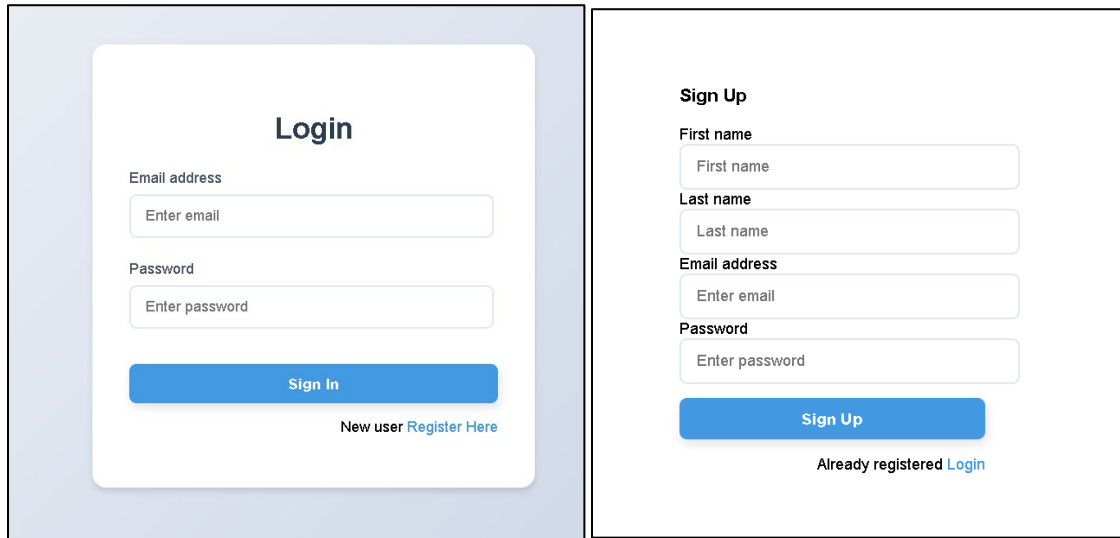
```
npm start
```

This will build the project and start the development server. It would open up a new tab in your browser, where you can interact with the project.

Usage

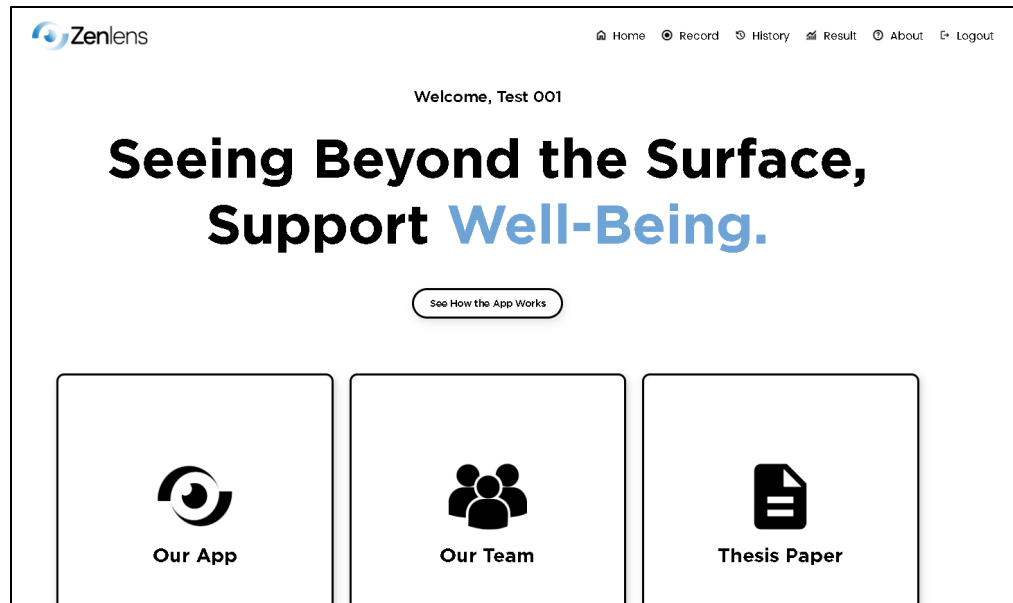
Now that both the frontend and backend are running, you can start using the system

1. Log in to the system using your credentials. If you are a first-time user, you can click the "Register Here" to redirect you into registering.



The image shows two side-by-side web forms. The left form is titled "Login" and contains fields for "Email address" (with placeholder "Enter email") and "Password" (with placeholder "Enter password"). Below these fields is a blue "Sign In" button and a link "New user Register Here". The right form is titled "Sign Up" and contains fields for "First name", "Last name", "Email address" (with placeholder "Enter email"), and "Password" (with placeholder "Enter password"). Below these fields is a blue "Sign Up" button and a link "Already registered Login".

2. After successfully logging in, you will be greeted with the home screen.

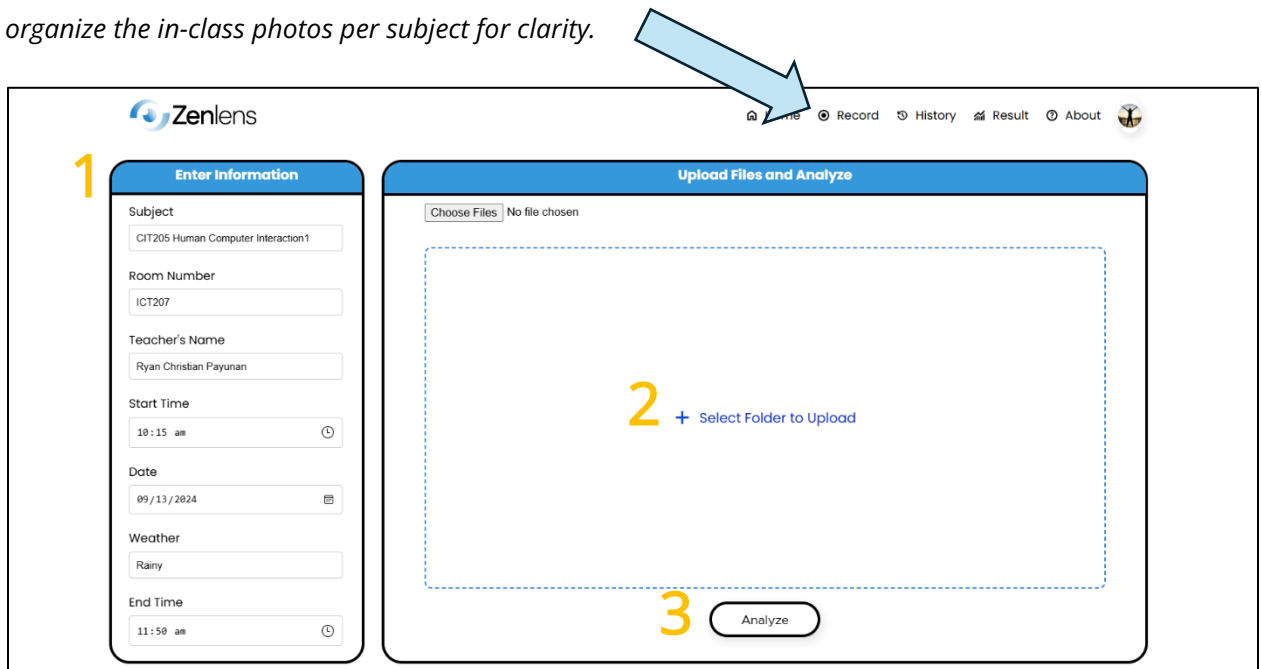


3. On the navigation bar, click on the **“Record”** button, which will lead you into this page. Fill out the necessary information to give the class context: the subject, room number, teacher’s name, the date, weather, and the start and end time of the subject.

After that, you can now select the folder (which contains the in-class photos taken during class hours) for emotion analysis. Then click **“Analyze”**.

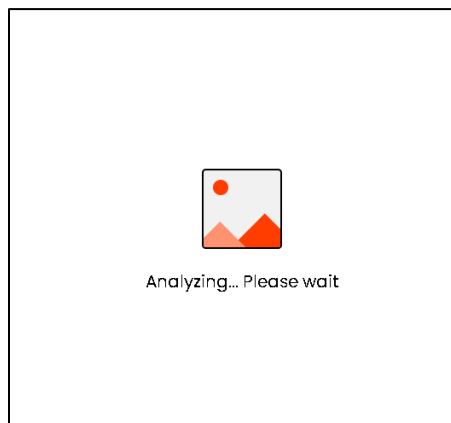
Note: It is recommended to upload a total of 60 photographs (for a 1.5-hour long class).

Note: The current system only supports uploading the whole folder. It is recommended to organize the in-class photos per subject for clarity.



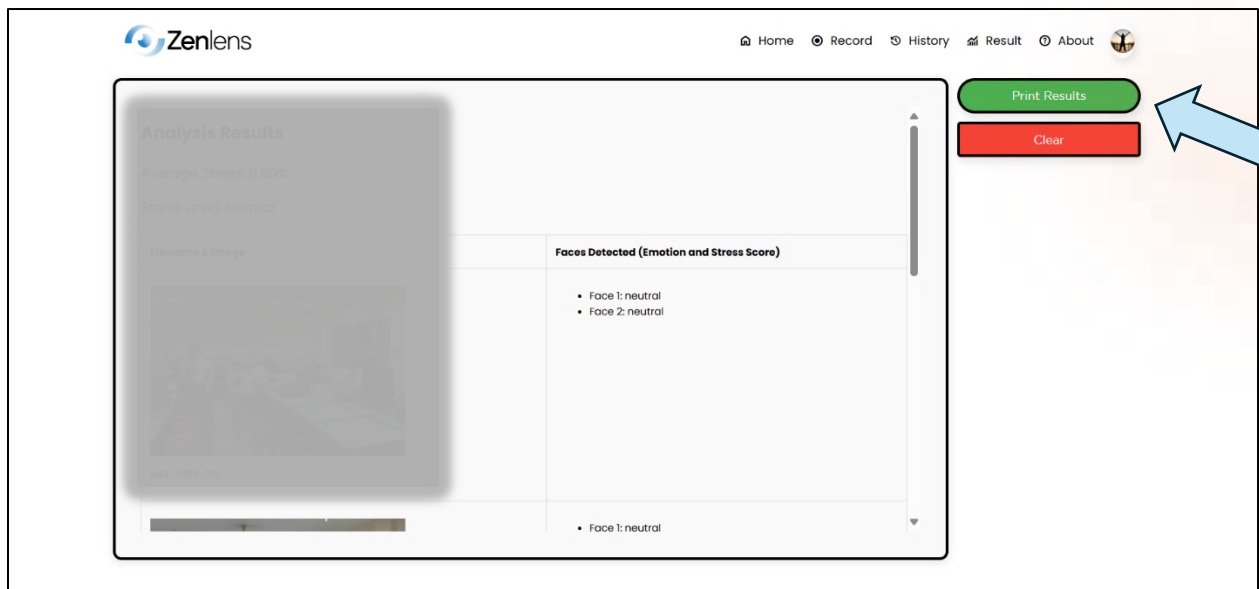
The screenshot shows the Zenlens web application interface. A blue arrow points from the 'Record' button in the navigation bar to the 'Enter Information' section. The interface is divided into two main panels. The left panel, titled 'Enter Information', contains form fields for Subject (CIT205 Human Computer Interaction1), Room Number (ICT207), Teacher's Name (Ryan Christian Payunan), Start Time (10:15 am), Date (09/13/2024), Weather (Rainy), and End Time (11:50 am). The right panel, titled 'Upload Files and Analyze', has a 'Choose Files' button, a 'No file chosen' status, a large dashed blue box for file selection, and an 'Analyze' button. Numbered annotations are present: a yellow '1' next to the 'Enter Information' header, a yellow '2' next to the '+ Select Folder to Upload' text, and a yellow '3' next to the 'Analyze' button.

4. During processing, it would take an average of 25-30 seconds for a series of 30 photos to finish processing (this would also depend on your computer’s performance.)

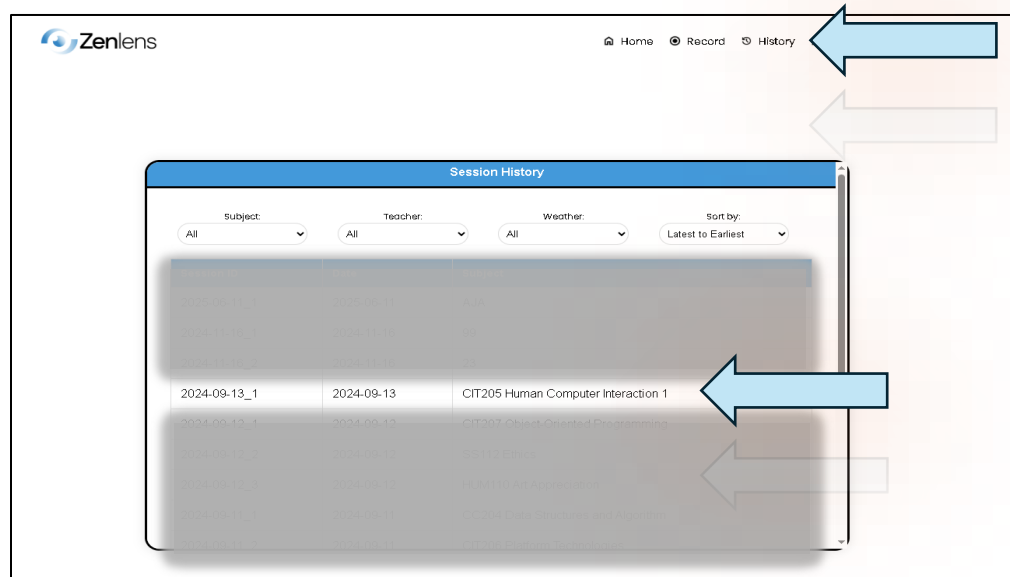


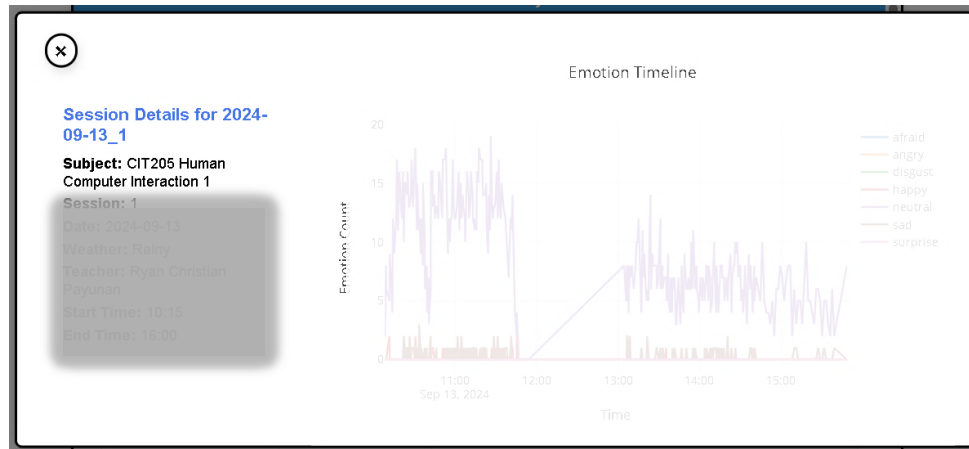
5. After that, results would be shown giving you the number of faces detected, emotions per face, and the corresponding stress level. You can save the file in PDF format by clicking the **"Print Results"** button.

Note: The following parts were blurred out for confidentiality purposes.



6. You can also view the monitoring session records by clicking the **"History"**. Moreover, a graph detailing the emotions progression over time is provided by clicking on a specific session. *Note: The following photo was blurred for confidentiality purposes.*





7. If you want to see the data grouped into **“Daily”** or **“Weekly”**, head over by clicking on the **“Results”** button.

The **“Weekly Data”** contains the recommendations on how to manage the stress given the stress level, which is powered by OpenAI’s gpt-3.5-turbo model.



Home Record History **Result** About Logout

Daily Data	
Daily Data ▾	
Date	
2024-09-09	View Details
2024-09-10	View Details
2024-09-11	View Details
2024-09-12	View Details
2024-09-13	View Details
2024-11-16	View Details
2024-05-02	View Details
2025-06-11	View Details

Weekly Data	
Weekly Data ▾	
Week	
2024-09-09 - 2024-09-13	View Details
2024-11-14 - 2024-11-14	View Details
2025-05-31 - 2025-05-31	View Details

Troubleshooting

An `ERROR:services.stress_utils:OpenAI AP call failed with status code 429:` appeared on the terminal.

- Make sure you have enough credit limits (paid) in your OpenAI account to get the recommendations. The analysis and recording of the system won't be affected, however, a text that says "No recommendation available for the selected week" would appear on the weekly data on the results page.

The uploaded photos are not processed.

- Make sure you have downloaded the model and saved it into the server folder.

User cannot see the sessions on the "History" and "Results" pages.

- Make sure you ran the flask server by following the instructions on the *Installation Step 7*.

Frequently Asked Questions (FAQs)

What is ZenLens?

- ZenLens is a user-friendly application designed to monitor emotional trends and provide insights using advanced AI technology.

Who can use ZenLens?

- ZenLens is designed for IT professionals, teachers, psychiatrists, and guidance counselors to assess emotional well-being.

How does ZenLens collect data?

- ZenLens uses pre-collected data to analyze emotions and trends without requiring any real-time input from users.

Is ZenLens secure?

- Yes, ZenLens follows strict data security and privacy guidelines to ensure user information is protected.

How does the user testing process work?

- Participants are provided with a pre-collected dataset and asked to interact with the system, evaluating its usability and features.

Can I provide feedback on ZenLens?

- Absolutely! Feedback from users is highly valued and helps us improve the application. You can find the team's contact details on the next section.

Is there a support team for ZenLens?"

- Yes, our support team is available to assist users with any questions or issues they may encounter. You can find the team's contact details on the next section.

What are the benefits of using ZenLens?

- ZenLens provides deep insights into emotional trends, helping professionals better understand and support their users.

Contact Details

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Disclaimer:

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